

SRKW-QERM514_LMM

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Fine Scale Diet Variability in Southern resident killer whales (*Orcinus orca ater*)

Southern resident killer whales (*Orcinus orca ater*; SRKW), with just 74 members of the population, have been listed as Endangered Species under the U.S. Endangered Species Act (ESA) since 2005 (Krahn et al., 2004). They occupy the inland and offshore waters of the Pacific Northwest, with ranges spanning from California to Southern Alaska; critical habitat for SRKWs includes Puget Sound, found in Washington state (Krahn et al., 2004). The main threats to SRKW are a lack of prey, vessel disturbance, contaminants (especially persistent organic pollutants (POPs)), and inbreeding (Switzer et al., 2026; Van Cise et al., 2024). Further, nutritional stress can work synergistically to amplify the effects of each stressor (Switzer et al., 2026). SRKW diets are culturally learned and believed to comprise almost entirely (>80%) of Chinook salmon (*Oncorhynchus tshawytscha*) (Van Cise et al., 2024). Many Chinook salmon populations in the Puget Sound region are also listed as threatened, endangered, or critical under the Endangered Species Act (Ford et al., 2011). For this project, I therefore aim to use data generated by genetic metabarcoding of the 16S region of mitochondrial DNA SRKW fecal samples collected from 2007 to 2025 (n=nearly 500) in Washington state and surrounding waters to describe any spatial, temporal, and/or demographic (e.g. by sex, pod, age class, etc.) changes therein. Library preparation was conducted by the San Diego Zoo and Wildlife Alliance (Escondido, CA).

Research Question: What is the fine-scale diet variability in SRKW (specifically, how does diet vary across pods, year, and season for SRKW)?

Hypotheses and Expectations:

hypothesis/hypotheses you're addressing; predictions from hypotheses Alternative Hypothesis 1: There will be variability in SRKW diet across pods (J, K, and L) Alternative Hypothesis 2: There will be variability in SRKW diet across year Alternative Hypothesis 3: There will be variability in SRKW diet across julian day (season)

Background about these data

Data generated by genetic metabarcoding of the 16S region of mitochondrial DNA SRKW fecal samples collected from 2007 to 2025 in Washington state and surrounding waters.

```
# Loads in output from DADA2 + filtered seqtab.nochim and samdf from rownames-match.r
load("replicates-contaminated.Rdata")

# Loads in species relative abundance x sample table
df <- read.csv("./Deliverables/ALL/SRKW_relative_speciesxsamples.csv")
```

Because of the nature of metabarcoding, absolute abundance cannot be determined from the methods. We therefore transform the raw counts into relative abundance data, which is reflected in df.

```

#?select()

# Messes with row and column names in samdf_filt for ease
samdf_filt <- samdf_filt %>%
  rename(Sample_name2 = Sample_name)%>%
  rownames_to_column("Sample_name")

# Selects the column for relative abundance of Chinook per sample
# renames column X to match samdf_filt
head(df)

```

```

##           X Cottus.asper Oncorhynchus.tshawytscha Oncorhynchus.keta
## 1      F21SEP11-01B-S1           0           0.068           0.36
## 2      F22MAY30-05A-S2           0           0.989           0.00
## 3      F22MAY30-05B-S3           0           0.992           0.00
## 4      F22SEP06-01C-S4           0           1.000           0.00
## 5 10Nov2022-3-Post-S5           0           0.953           0.00
## 6      F23JUL01-03A-S6           0           1.000           0.00
## Oncorhynchus.kisutch Ophiodon.elongatus Oncorhynchus.nerka Raja.binoculata
## 1           0.559           0.000           0           0
## 2           0.000           0.000           0           0
## 3           0.000           0.000           0           0
## 4           0.000           0.000           0           0
## 5           0.000           0.042           0           0
## 6           0.000           0.000           0           0
## Oncorhynchus.mykiss Hydrolagus.colliei Anoplopoma.fimbria
## 1           0.000           0           0
## 2           0.008           0           0
## 3           0.003           0           0
## 4           0.000           0           0
## 5           0.005           0           0
## 6           0.000           0           0
## Hippoglossus.stenolepis Atheresthes.stomias Raja.rhina NA.
## 1           0           0           0 0.013
## 2           0           0           0 0.003
## 3           0           0           0 0.005
## 4           0           0           0 0.000
## 5           0           0           0 0.001
## 6           0           0           0 0.000
## Citharichthys.stigmaeus NA.1 Oncorhynchus.gorbuscha Salmo.salar
## 1           0 0           0           0
## 2           0 0           0           0
## 3           0 0           0           0
## 4           0 0           0           0
## 5           0 0           0           0
## 6           0 0           0           0
## Clupea.pallasii Gadus.chalcogrammus Citharichthys.sordidus Liparis.pulchellus
## 1           0           0           0           0
## 2           0           0           0           0
## 3           0           0           0           0
## 4           0           0           0           0
## 5           0           0           0           0
## 6           0           0           0           0

```

```

##   Zaprora.silenus Carcharodon.carcharias Myoxocephalus.polyacanthocephalus
## 1           0                0                0
## 2           0                0                0
## 3           0                0                0
## 4           0                0                0
## 5           0                0                0
## 6           0                0                0
##   Cymatogaster.aggregata Glyptocephalus.zachirus NA.2 Ammodytes.hexapterus NA.3
## 1           0                0  0                0  0
## 2           0                0  0                0  0
## 3           0                0  0                0  0
## 4           0                0  0                0  0
## 5           0                0  0                0  0
## 6           0                0  0                0  0
##   NA.4 Liparis.florae NA.5 Microstomus.pacificus Anarrhichthys.ocellatus NA.6
## 1  0           0  0                0                0  0
## 2  0           0  0                0                0  0
## 3  0           0  0                0                0  0
## 4  0           0  0                0                0  0
## 5  0           0  0                0                0  0
## 6  0           0  0                0                0  0
##   NA.7 Diaphus.theta Gasterosteus.aculeatus Engraulis.mordax
## 1  0           0                0                0
## 2  0           0                0                0
## 3  0           0                0                0
## 4  0           0                0                0
## 5  0           0                0                0
## 6  0           0                0                0
##   Liparis.sp..BOLD.AAB4898 Squalus.acanthias NA.8 Merluccius.productus
## 1           0                0  0                0
## 2           0                0  0                0
## 3           0                0  0                0
## 4           0                0  0                0
## 5           0                0  0                0
## 6           0                0  0                0
##   Thaleichthys.pacificus
## 1           0
## 2           0
## 3           0
## 4           0
## 5           0
## 6           0

```

```

df_chin <- df %>%
  dplyr::select(X, Oncorhynchus.tshawytscha)%>%
  dplyr::rename(Sample_name = X) %>%
  dplyr::rename(y = Oncorhynchus.tshawytscha)

# Adds metadata from samdf_filt to df_chin by Sample_name
# ?left_join()
df_chin <- left_join(df_chin, samdf_filt, by = "Sample_name")

# Cleans Pod column
df_chin$Pod[df_chin$Pod == "unknown"] <- "UNK"

```

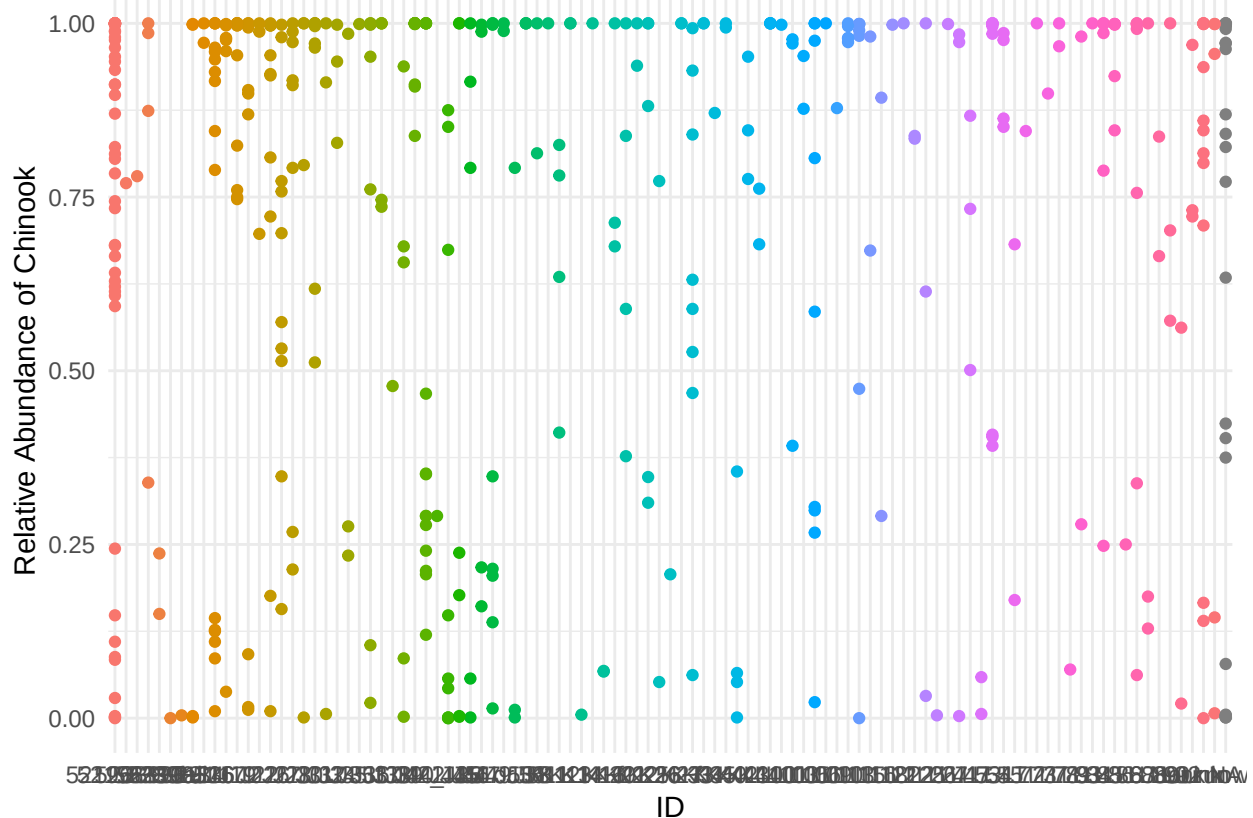
```
df_chin$Pod[is.na(df_chin$Pod)] <- "UNK"
df_chin$Pod[df_chin$Pod == "NA"] <- "UNK"
df_chin$Pod[df_chin$Pod == "J,K,L"] <- "UNK"
df_chin$Pod[df_chin$Pod == "J,L"] <- "UNK"
```

In df_chin, we have:

y = the percent of chinook in each sample (relative) ID = individual whale Year = the year the sample was collected Pod = the pod that the individual whale belongs to julian_day = converted from Day, Month, Year that the sample was collected; used as a continuous proxy for season

```
# Plots the relative abundance of Chinook per sample
p1 <- ggplot(df_chin, aes(ID, y, color = ID)) +
  geom_point() +
  ylab("Relative Abundance of Chinook") +
  theme_minimal() +
  theme(legend.position = "none")
```

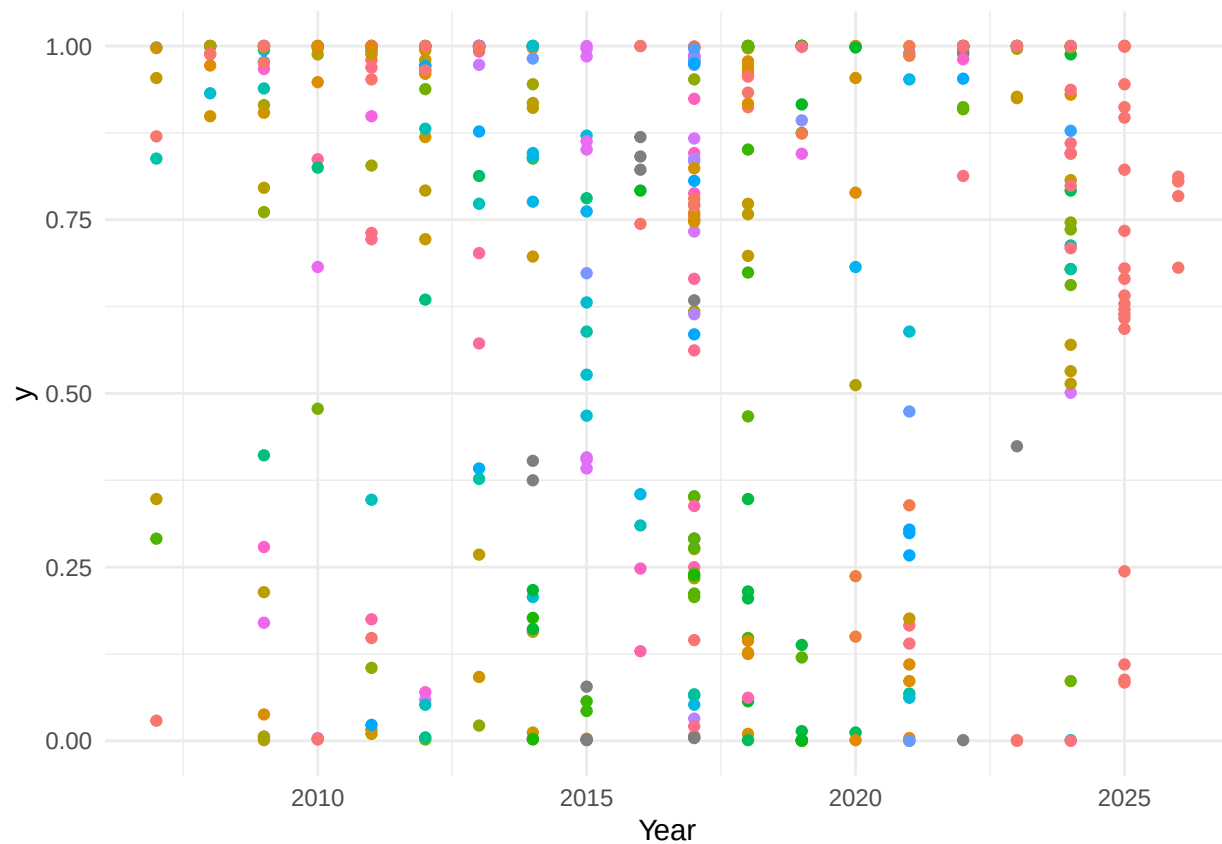
p1



```
# Plots the relative abundance of Chinook per year
p2 <- ggplot(df_chin, aes(Year, y, color = ID)) +
  geom_point() +
  theme_minimal() +
  theme(legend.position = "none")
```

p2

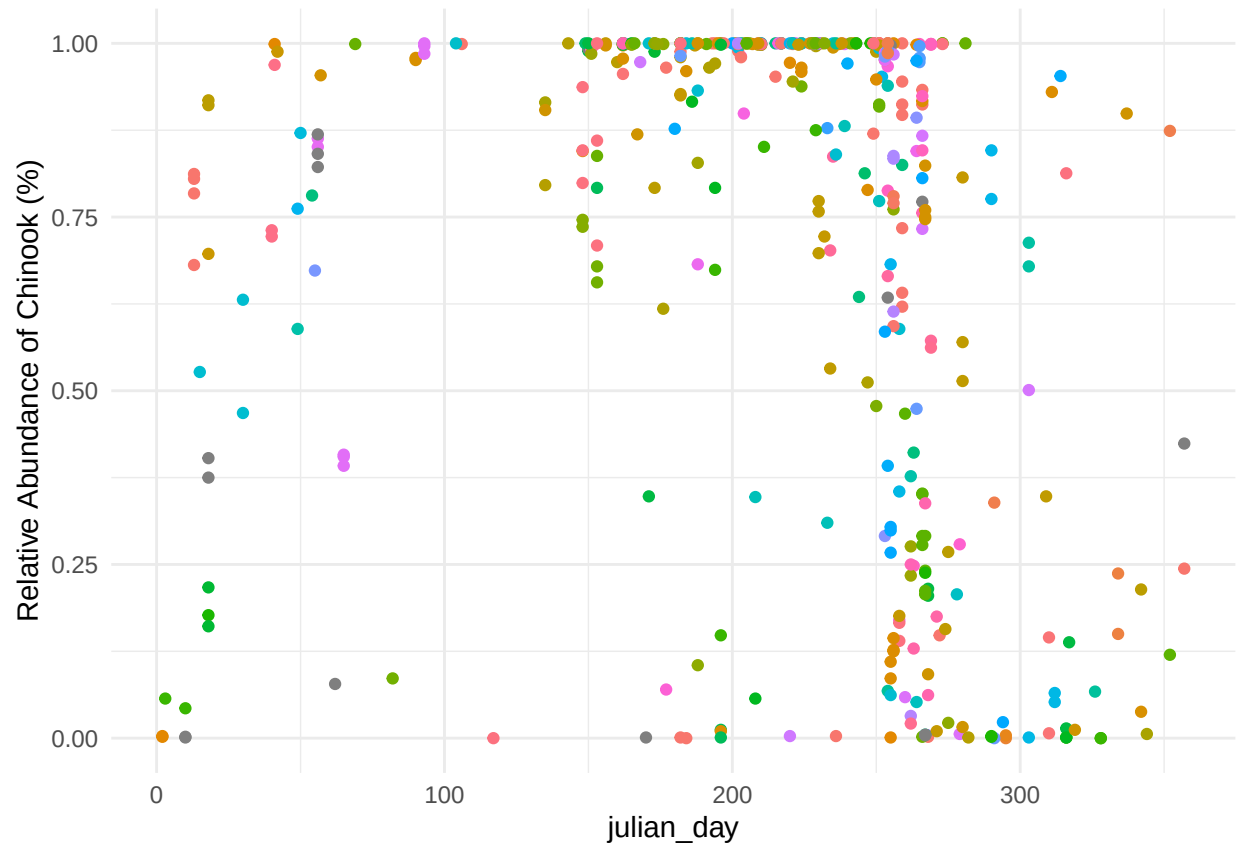
```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
# Plots the relative abundance of Chinook per season
p3 <- ggplot(df_chin, aes(julian_day, y, color = ID)) +
  geom_point() +
  ylab("Relative Abundance of Chinook (%)") +
  theme_minimal() +
  theme(legend.position = "none")
```

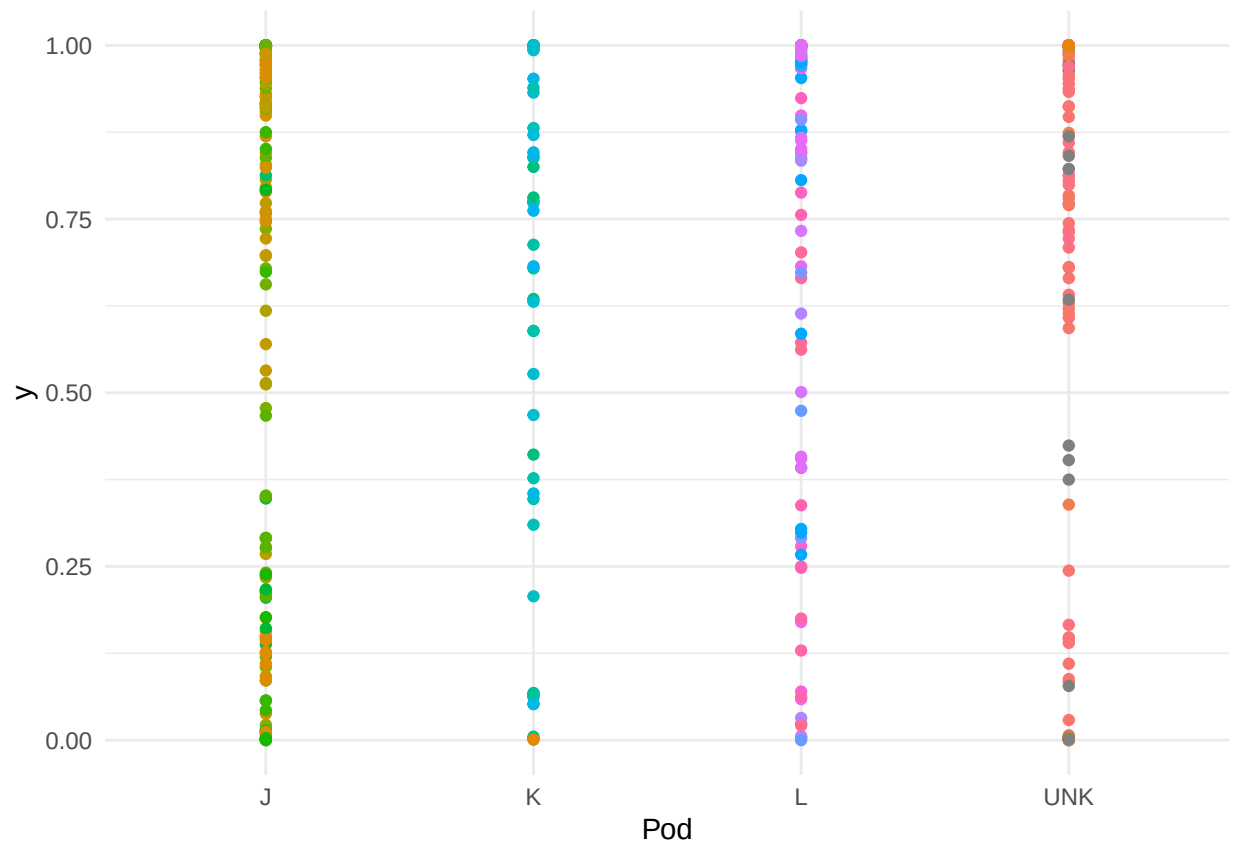
p3

```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
# Plots the relative abundance of Chinook per pod
p4 <- ggplot(df_chin, aes(Pod, y, color = ID)) +
  geom_point()+
  # ylab("Relative Abundance of Chinook (%)")+
  theme_minimal()+
  theme(legend.position = "none")
```

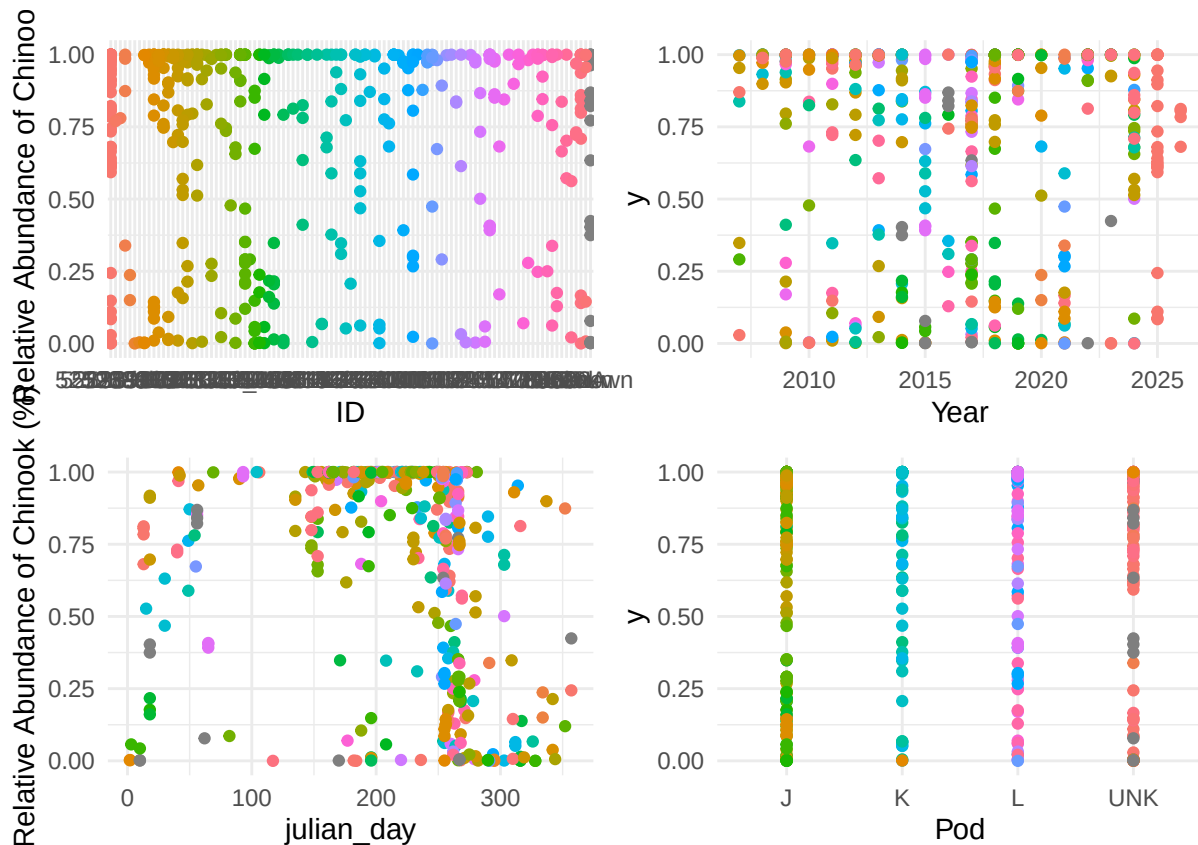
p4



```
(p1 + p2) / (p3 + p4)
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## ('geom_point()').
```



According to these plots, I see some possible differences across parameters. For individual, I see that there are many whales for which we have more than one sample. I also see a wide spread of percent of chinook found in the diet (0%-100% chinook). Because there is individual variation here, I want to make sure to capture that in the model. Because each sample from the same individual will not be independent (samples from one individual can be thought of as a “grouping”), I plan to include this as a random effect.

I don’t see a clear pattern for relative abundance of chinook in the diet by year.

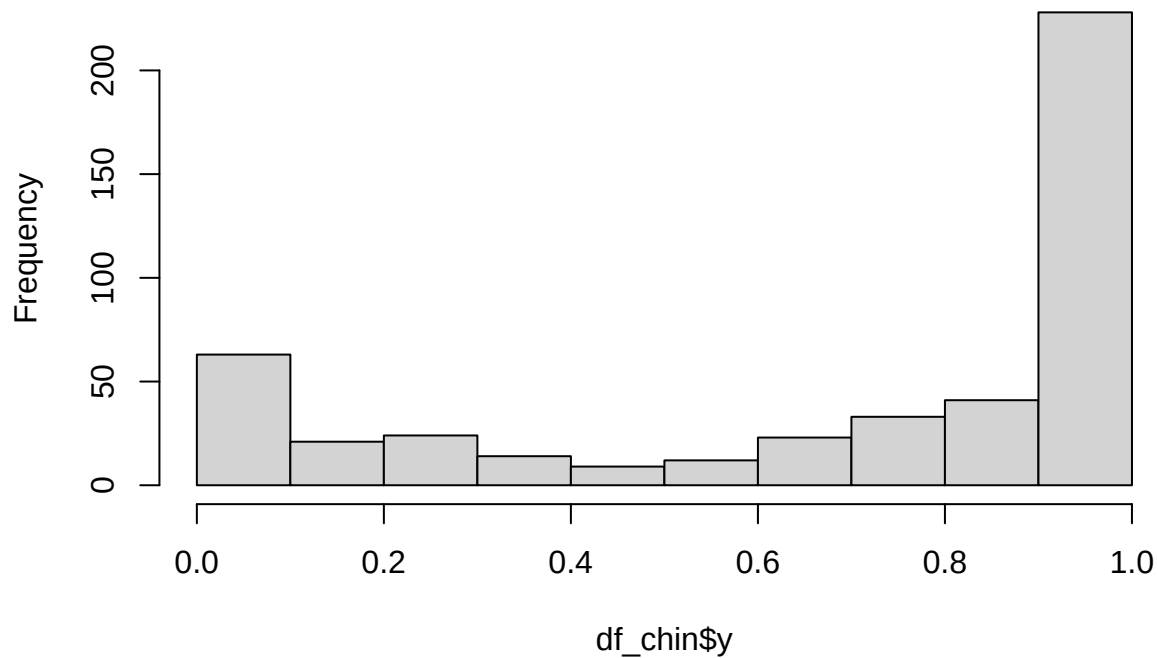
I do see some structure by Julian day, which can be used as a continuous proxy for season. Though it does appear like sampling effort was lower in the Spring compared to the Summer, Fall, and Winter, with sampling effort being the highest in September (Julian day ~70). I also see that individuals from different pods may be more prevalent in our sampling areas in certain seasons. For example, individuals from J pod seem to make up most of the available samples from the summer months, and samples from L pod may be more available in the Autumn. Interestingly, there may also be a higher percent of Chinook in the diet for L pod in Autumn vs J pod, but it is hard to make any real statements at this time.

Upon visual inspection, the data does not look normally distributed.

```
# Visual and statistical tests for normality

# Histogram of the relative abundance of chinook
# ?hist()
hist(df_chin$y) #this data definitely does not appear normal
```


Histogram of df_chin\$y



```
# Shapiro Wilkes test for normality

# makes chin in df_chin an numeric vector
x <- df_chin$y
range(x)

## [1] 0 1

# Runs Shapiro-Wilk normality test
## Null hypothesis: the data are normal
## Alt hypotheeiss: the data are not normal
## alpha = 0.5

shapiro.test(x) #p = 2.2e-16

##
## Shapiro-Wilk normality test
##
## data: x
## W = 0.77205, p-value < 2.2e-16
```

Using a Shapiro-Wilk test, this is corroborated. With a p value $\ll 0.05$, we reject the null and conclude that the data are not normally distributed.

This suggests that I will need more than just a simple linear model. Because of the distribution of this data, and because it is proportional data, I will try logit transforming my y and then using a linear mixed

model; the logit is a transformation that takes a probability p (a value between 0 and 1) and maps it to a continuous, unconstrained value spanning from $-\infty$ to $+\infty$

```
## I have some true 1's and some true 0's, this nudges them by 0.001
```

```
eps <- 0.001
```

```
df_chin$y_nudge <- pmin(pmax(df_chin$y, eps), 1 - eps)
```

```
# transforms chin column
```

```
# chin = percent of chinook in diet per sample
```

```
# y_logit = logit of the percent of chinook in diet per sample
```

```
df_chin <- df_chin %>%
```

```
  mutate(y_logit = log(y_nudge/(1-y_nudge)))
```

```
# Plots the relative abundance of Chinook per sample
```

```
p5 <- ggplot(df_chin, aes(ID, y_logit, color = ID)) +
```

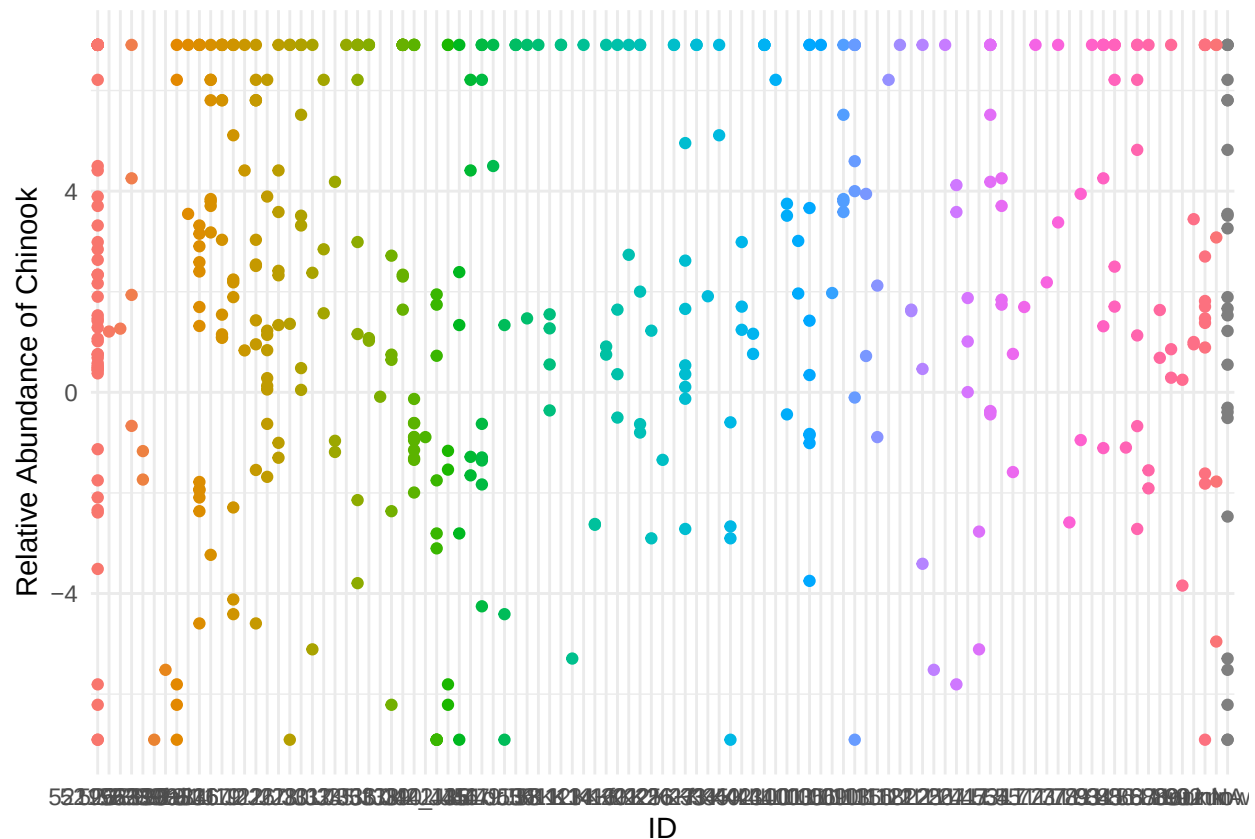
```
  geom_point() +
```

```
  ylab("Relative Abundance of Chinook") +
```

```
  theme_minimal() +
```

```
  theme(legend.position = "none")
```

p5



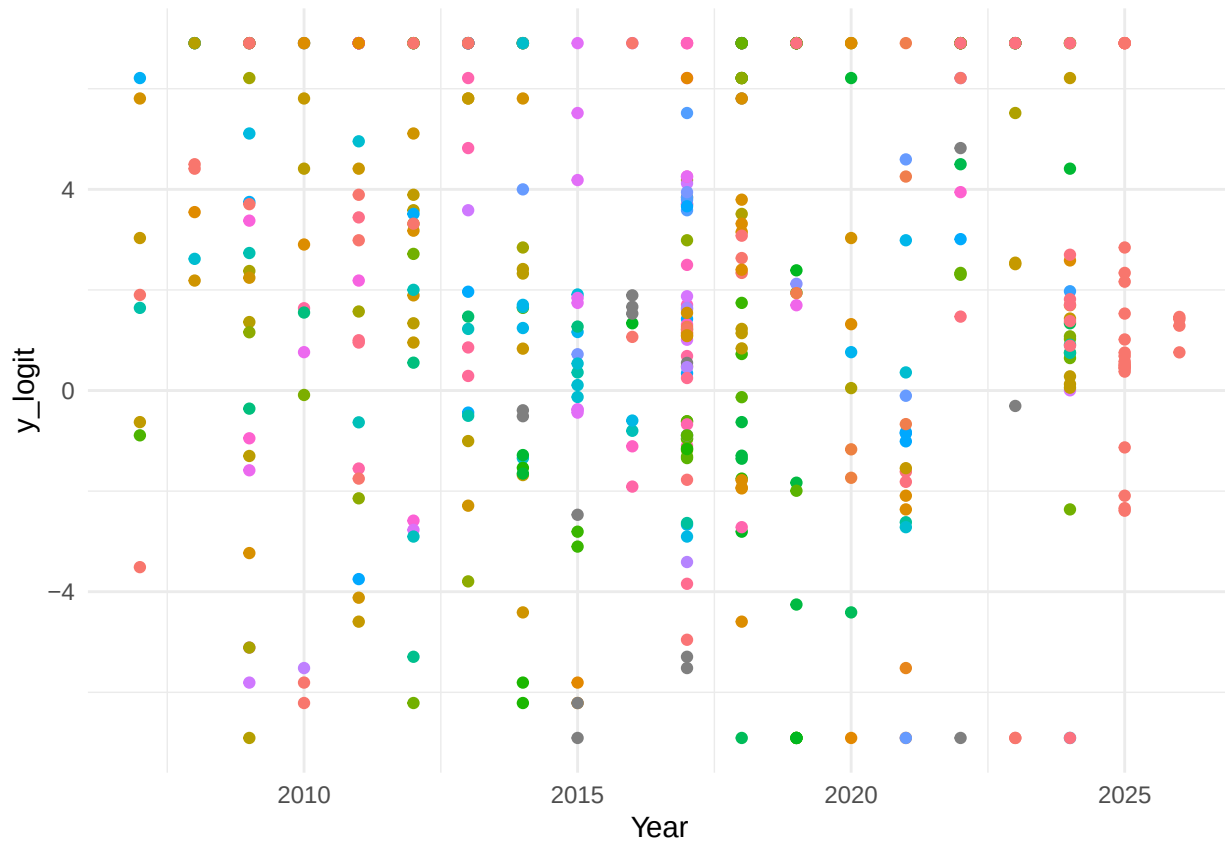
```
# Plots the relative abundance of Chinook per year
```

```
p6 <- ggplot(df_chin, aes(Year, y_logit, color = ID)) +
```

```
geom_point() +
theme_minimal()+
theme(legend.position = "none")
```

p6

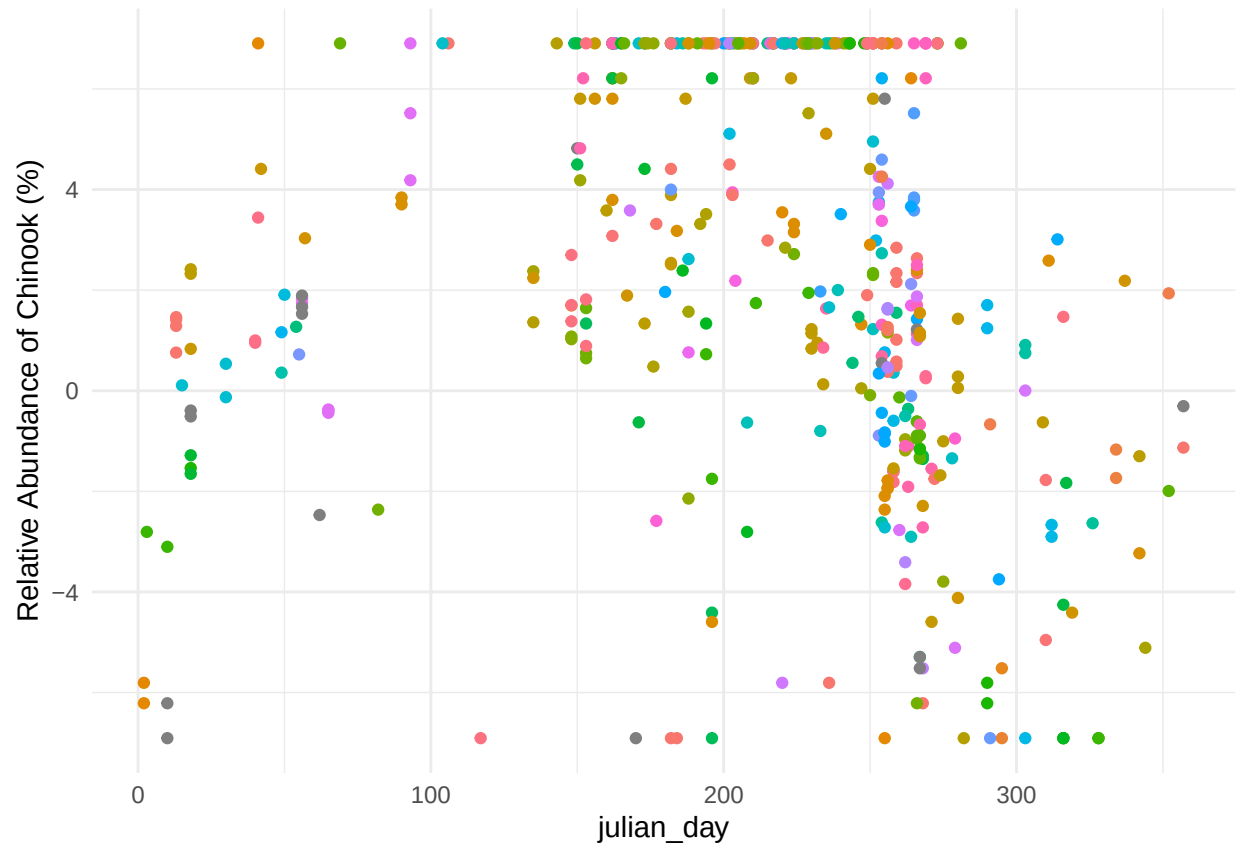
```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
# Plots the relative abundance of Chinook per season
p7 <- ggplot(df_chin, aes(julian_day, y_logit, color = ID)) +
  geom_point()+
  ylab("Relative Abundance of Chinook (%)")+
  theme_minimal() +
  theme(legend.position = "none")
```

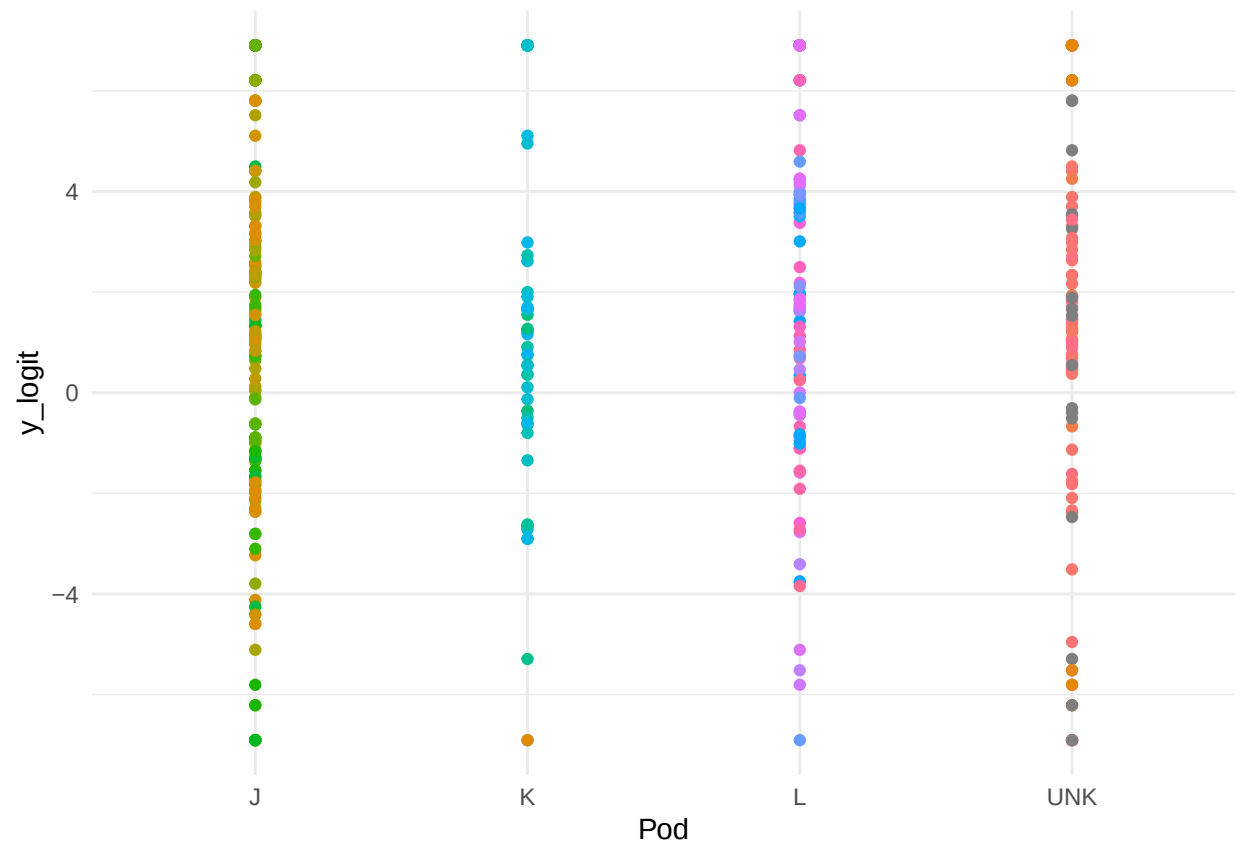
p7

```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
# Plots the relative abundance of Chinook per pod
p8 <- ggplot(df_chin, aes(Pod, y_logit, color = ID)) +
  geom_point()+
  # ylab("Relative Abundance of Chinook (%)")+
  theme_minimal()+
  theme(legend.position = "none")

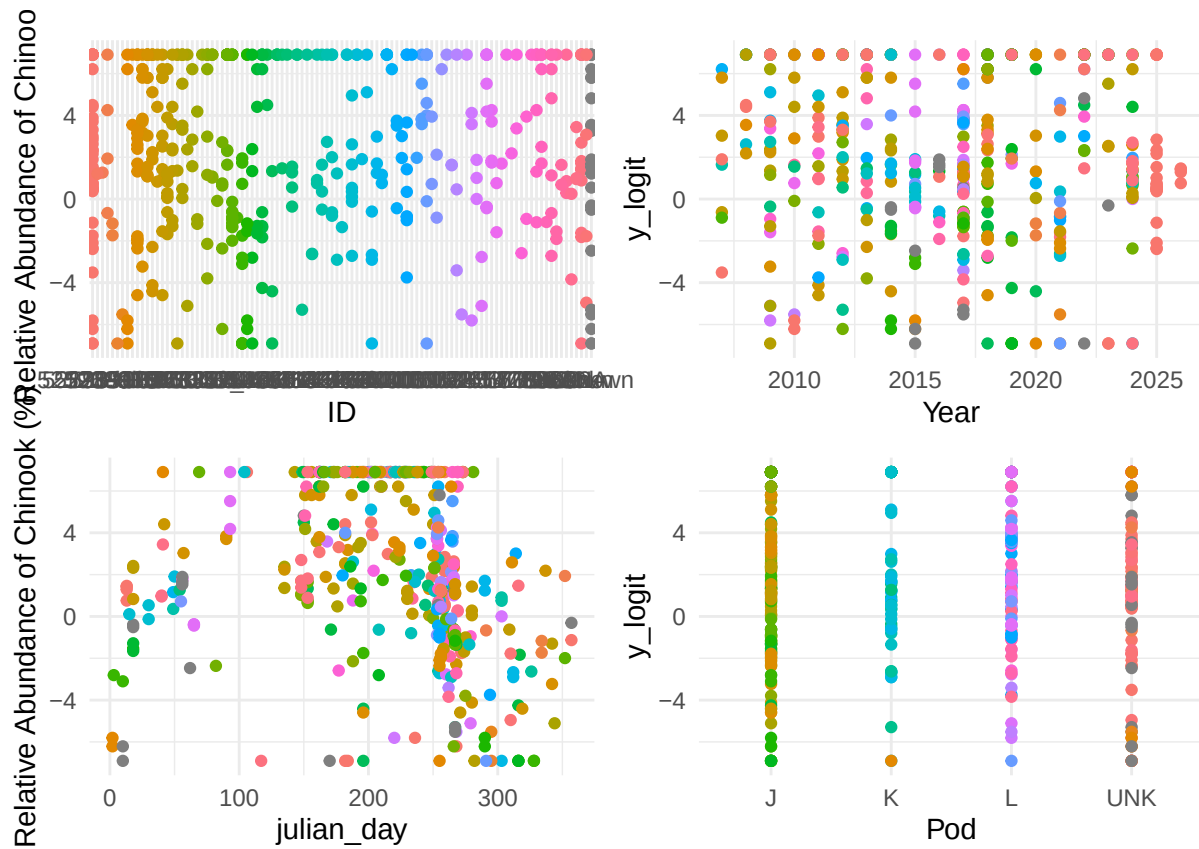
p8
```



```
(p5 + p6) / (p7 + p8)
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## ('geom_point()').
```

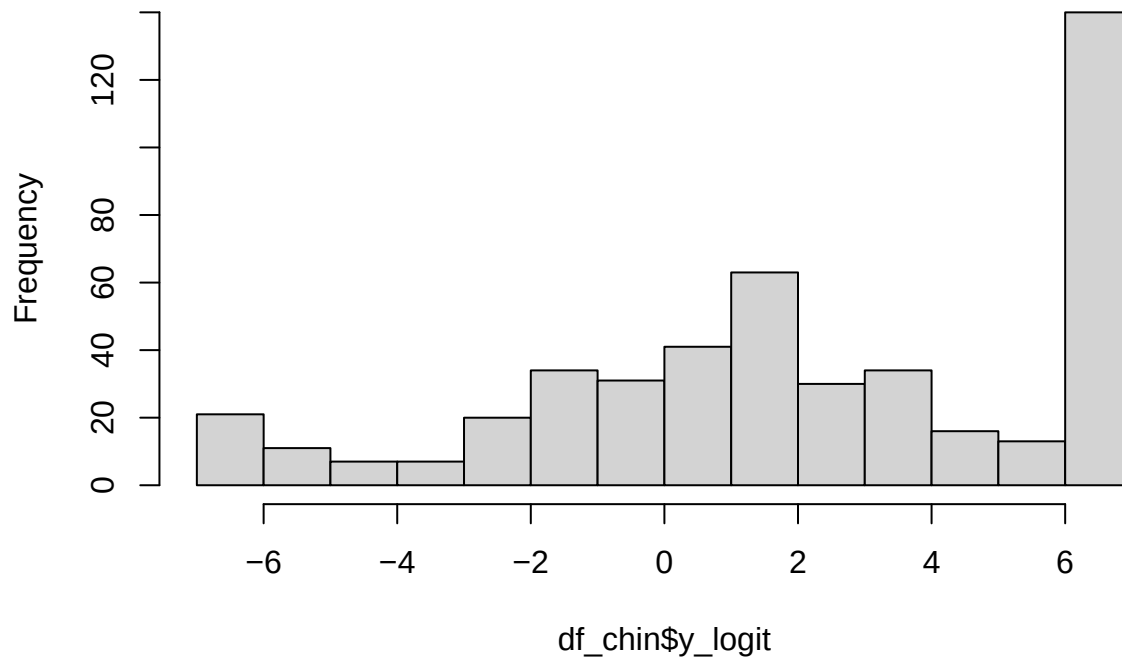


This still does not look that much different. To test for normality now:

```
# Visual and statistical tests for normality

# Histogram of the relative abundance of chinook
# ?hist()
hist(df_chin$y_logit) #this data definitely does not appear normal
```

Histogram of df_chin\$y_logit



```
# Shapiro Wilkes test for normality  
  
# makes chin in df_chin an numeric vector  
x <- df_chin$y_logit  
range(x)
```

```
## [1] -6.906755  6.906755
```

```
# Runs Shapiro-Wilk normality test  
## Null hypothesis: the data are normal  
## Alt hypotheeiss: the data are not normal  
## alpha = 0.5  
  
shapiro.test(x) #p = 1.016e-15
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  x  
## W = 0.91336, p-value = 1.016e-15
```

These data are still not normally distributed. However, the residuals could be normally distributed.

Thus, I will proceed exploring the fit of linear mixed effect models. Specifically, I will look for (1) residuals that are normal and homoscedastic, (2) violations of other LMM assumptions (linearity via QQ plot), etc.,

and (3) examine the fixed effects. I need a mixed effect model because ID should be a random effect: I expect that samples collected from the same individual are more similar than samples collected from different individuals.

I will fit four models and compare them: 1. all three fixed effects + my random effect 2. only pod as a fixed effect + my random effect 3. only year as a fixed effect + my random effect 4. only julian day as a fixed effect + my random effect.

I expect to see that the best model is Julian day + the random effect. This is consistent with anecdotal information from field observations (etc.), where SRKWs seem to eat more chinook in the summer months and supplement their diet with lingcod and other species of salmon in other months.

However, there is also anecdotal evidence out there that suggests that different pods may eat chinook salmon at different proportions, so I will be looking for evidence of that, given the random effect of individual.

1. all three fixed effects + my random effect

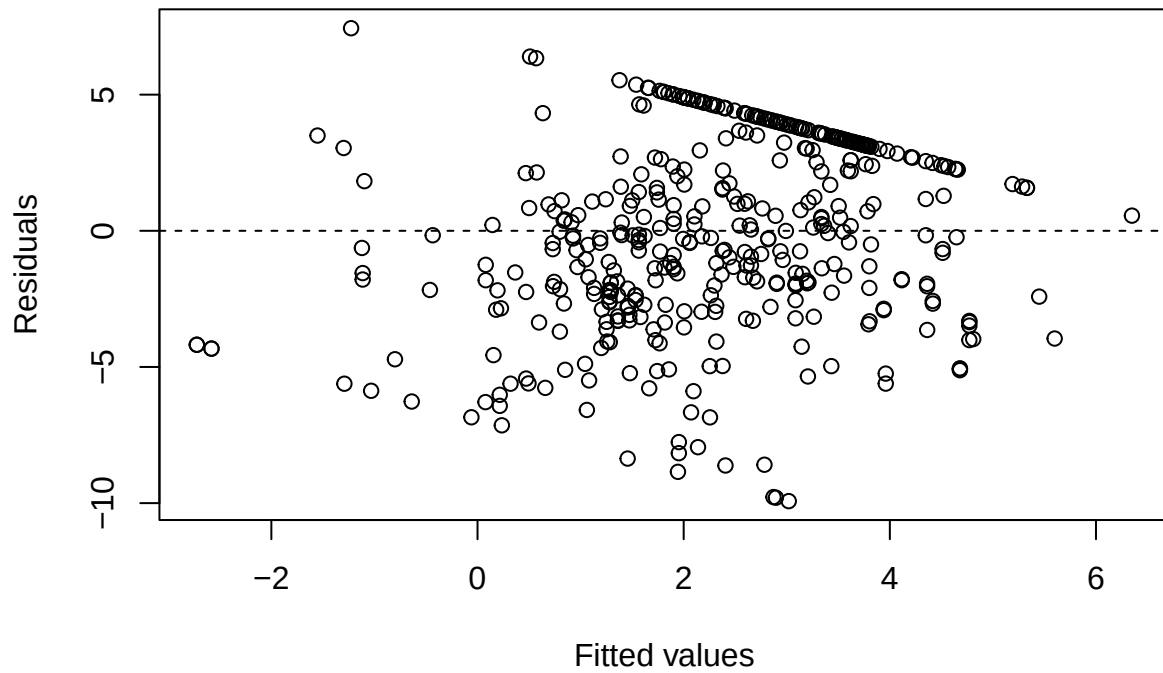
$$y_i = \beta_0 - \beta_1 \text{Year}_i + \beta_2 \text{julian}_i + \beta_3 \text{Pod}_i + \delta_{ID_i} + \epsilon_i \epsilon_i \sim N(0, \sigma_\epsilon^2) \delta_i \sim N(0, \sigma_\delta^2)$$

```
# runs the lm with all the fixes effects and random effect
allwRE_mod <- lmer(y_logit ~ Year + julian_day + Pod + (1 | ID), data = df_chin)
summary(allwRE_mod)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: y_logit ~ Year + julian_day + Pod + (1 | ID)
## Data: df_chin
##
## REML criterion at convergence: 2374.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.74444 -0.65340 -0.00526  0.86849  2.05681
##
## Random effects:
## Groups Name Variance Std.Dev.
## ID      (Intercept) 2.609 1.615
## Residual 13.081 3.617
## Number of obs: 428, groups: ID, 99
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 86.651643 85.684218 1.011
## Year -0.040600 0.042501 -0.955
## julian_day -0.011827 0.002626 -4.503
## PodK -0.368129 0.762908 -0.483
## PodL 0.495567 0.623967 0.794
## PodUNK -0.971673 0.900001 -1.080
##
## Correlation of Fixed Effects:
## (Intr) Year jln_dy PodK PodL
## Year -1.000
## julian_day 0.024 -0.031
## PodK -0.108 0.106 -0.040
## PodL -0.109 0.106 -0.087 0.367
## PodUNK 0.087 -0.089 0.031 0.241 0.282
```

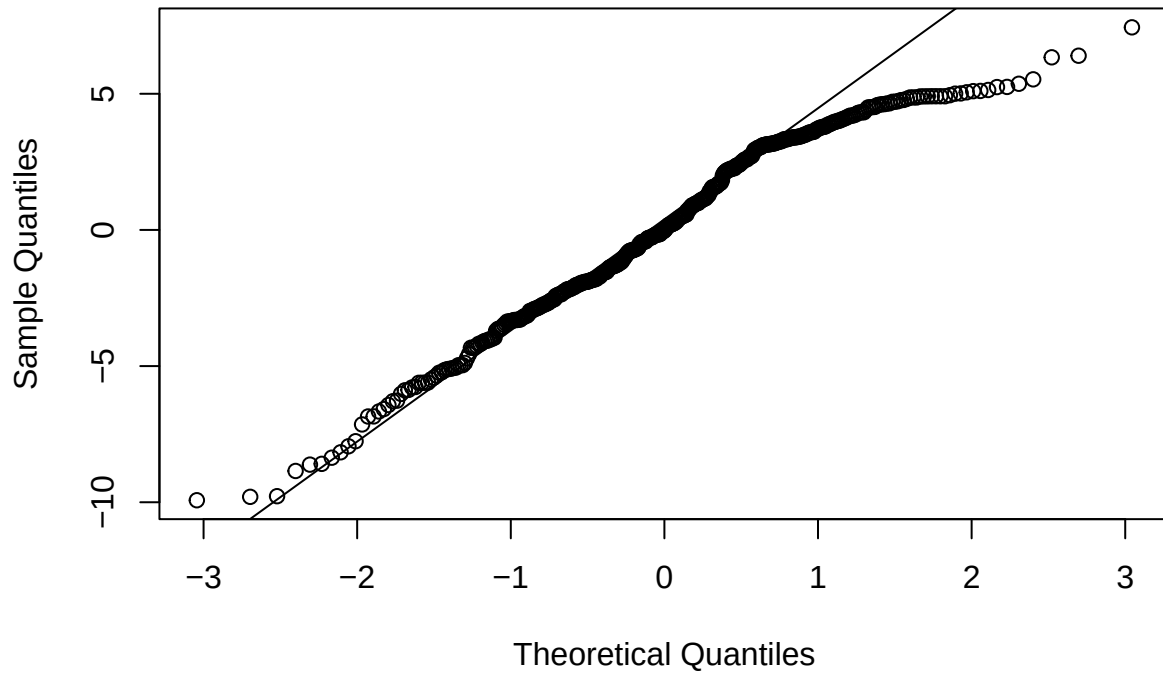


```
# Residuals vs fitted
plot(fitted(allwRE_mod), resid(allwRE_mod),
     xlab = "Fitted values", ylab = "Residuals")
abline(h = 0, lty = 2)
```

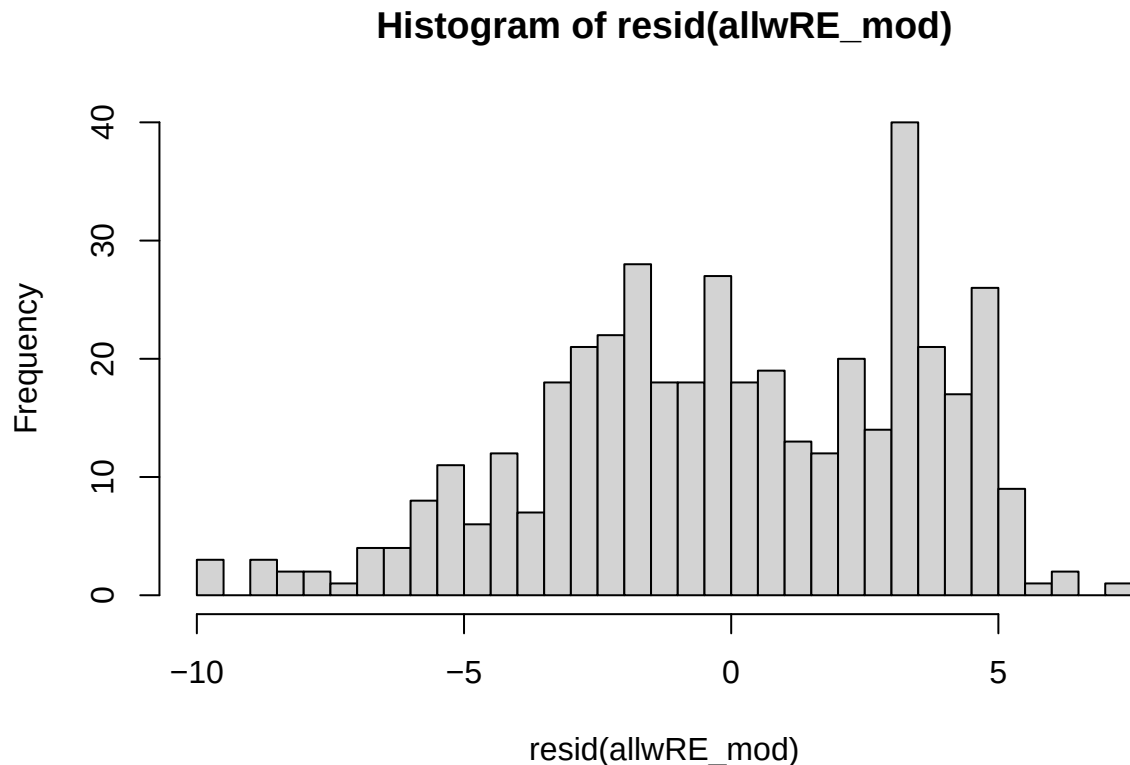


```
# QQ-plot of residuals
qqnorm(resid(allwRE_mod))
qqline(resid(allwRE_mod))
```

Normal Q-Q Plot



```
# Histogram of residuals  
hist(resid(allwRE_mod), breaks = 30)
```



All effect sizes are small and none of the predictors are significant.

Checking our assumptions, there is some structure in the residual vs fitted plot. There is a clear line of points at the top right of the plot. This is a bit concerning for me for sure. For the QQ plot, I am not surprised to see heavy deviation from the line in the top right. We saw from our histogram that our data was heavy-tailed, particularly in the higher quantiles, which is reflected in the histogram.

```
performance::r2(allwRE_mod)
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.216
##   Marginal R2: 0.059
```

The marginal R^2 is 0.059, meaning that this model is a poor fit (accounts for <6% of the variation in the data). I report the marginal based on the 5/15 lab conversation.

Though there are some issues with this model, I'm still going to examine the fixed effects to see what combination of fixed effects best fits the data.

Also, there are a few points (412, 397, and 376 especially) that may have high leverage, and probably need to be looked at a little closer.

```
infl <- influence(allwRE_mod, obs = TRUE)      # or group = "ID"
cd   <- cooks.distance(infl)
ord  <- order(cd, decreasing = TRUE)
head(ord)                                     # row numbers of most influential obs
```

```
## [1] 444 443 364 412 393 413
```

```
cd[ord[1:6]]          # their Cook's distances
```

```
## [1] 0.03727390 0.03377404 0.03250634 0.02724180 0.02610150 0.01735461
```

```
df_chin[ord[1:6], ]  # inspect those rows
```

```
##      Sample_name      y Sample_no      Sample_name2      ID Pre.Post_hormone
## 444 SRR27755031 0.002      NA 2015Jan02-A-reex_S24      fail      <NA>
## 443 SRR27755032 0.003      NA 2015Jan02-A_S1      fail      <NA>
## 364 SRR27755003 0.998      NA 52196-161_S42      fail      <NA>
## 412 SRR27755250 0.000      NA 52390-35_S29 52390-34      <NA>
## 393 SRR27755147 0.001      NA 52389-16_S11      fail      <NA>
## 413 SRR27755269 0.004      NA 52390-36_S30 52390-36      <NA>
##      Year Month Day Pod FieldID X LabID Sample.ID      Sex pedPod
## 444 2015 Jan 2 UNK <NA> 271 52196-038 2015Jan10-02      fail <NA>
## 443 2015 Jan 2 UNK <NA> 270 52196-038 2015Jan10-02      fail <NA>
## 364 2017 Sep 21 UNK <NA> 59 52196-161 2017SEP21-06      fail <NA>
## 412 2021 Oct 22 UNK <NA> 111 52390-35 2021Oct22-04b      Female L
## 393 2020 Sep 11 K <NA> 90 52389-16 2020Sep11-03 Undetermined <NA>
## 413 2021 Oct 22 UNK <NA> 112 52390-36 2021Oct22-05b      Male L
##      Region MiSeq.run Population latitudeN longitudeW      X.1
## 444 WA 2016 SRKW 47.81054 122.4193 47.81054_122.41929
## 443 WA 2016 SRKW 47.81054 122.4193 47.81054_122.41929
## 364 WA 2019 SRKW 48.97225 123.1171 48.97225_123.11713
## 412 WA 2021P2 SRKW 48.46862 124.6941 48.468623_124.69409
## 393 WA 2021P2 SRKW 48.55678 123.2309 48.556775_123.230931
## 413 WA 2021P2 SRKW 48.46862 124.6941 48.468623_124.69409
##      dom.sp      prey50 TopRel      Run yy mon dd
## 444 Raja binoculata Raja binoculata <NA> SRR27755031 <NA> <NA> <NA>
## 443 Raja binoculata Raja binoculata <NA> SRR27755032 <NA> <NA> <NA>
## 364 chinook chinook <NA> SRR27755003 <NA> <NA> <NA>
## 412 sablefish sablefish PO SRR27755250 <NA> <NA> <NA>
## 393 coho coho <NA> SRR27755147 <NA> <NA> <NA>
## 413 <NA> None PO SRR27755269 <NA> <NA> <NA>
##      year_from_name month_from_name day_from_name      date julian_day
## 444 NA <NA> NA 2015-01-02 2
## 443 NA <NA> NA 2015-01-02 2
## 364 NA <NA> NA 2017-09-21 264
## 412 NA <NA> NA 2021-10-22 295
## 393 NA <NA> NA 2020-09-11 255
## 413 NA <NA> NA 2021-10-22 295
##      pod_from_id y_nudge y_logit
## 444 <NA> 0.002 -6.212606
## 443 <NA> 0.003 -5.806138
## 364 <NA> 0.998 6.212606
## 412 <NA> 0.001 -6.906755
## 393 <NA> 0.001 -6.906755
## 413 <NA> 0.004 -5.517453
```

Cook's distances of $> 4/n$ may need a further look. $4/468 = 0.0085$. That's a pretty small number, and my influential points have a $CD > 0.0085$. However, they are all < 0.5 , and $\ll 1$, so I am not worried about high leverage points.

2. only pod as a fixed effect + my random effect

$$y_i = \beta_0 - \beta_1 Pod_i + \delta_{ID_i} + \epsilon_i \epsilon_i \sim N(0, \sigma_\epsilon^2) \delta_i \sim N(0, \sigma_\delta^2)$$

```
podwRE_mod <- lmer(y_logit ~ Pod + (1 | ID), data = df_chin)
summary(podwRE_mod)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: y_logit ~ Pod + (1 | ID)
## Data: df_chin
##
## REML criterion at convergence: 2462.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.68088 -0.67780  0.00332  0.83634  1.87667
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## ID       (Intercept)  2.593    1.610
## Residual                13.529    3.678
## Number of obs: 444, groups: ID, 100
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)   2.2915     0.4094   5.597
## PodK          -0.4022     0.7602  -0.529
## PodL           0.3511     0.6210   0.565
## PodUNK        -1.0474     0.8987  -1.165
##
## Correlation of Fixed Effects:
##      (Intr) PodK   PodL
## PodK   -0.538
## PodL   -0.659  0.355
## PodUNK -0.447  0.252  0.295
```

All effect sizes are similiary small and non-significant.

```
performance::r2(podwRE_mod)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.173
## Marginal R2: 0.014
```

The marginal R^2 is 0.014. This model is worse than all the fixed effects included.

3. only year as a fixed effect + my random effect

$$y_i = \beta_0 - \beta_1 Year_i + \delta_{ID_i} + \epsilon_i \epsilon_i \sim N(0, \sigma_\epsilon^2) \delta_i \sim N(0, \sigma_\delta^2)$$

```
YearwRE_mod <- lmer(y_logit ~ Year + (1 | ID), data = df_chin)
summary(YearwRE_mod)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: y_logit ~ Year + (1 | ID)
## Data: df_chin
##
## REML criterion at convergence: 2471.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.5727 -0.6758 -0.0140  0.8798  1.6523
##
## Random effects:
## Groups Name Variance Std.Dev.
## ID      (Intercept) 2.449  1.565
## Residual      13.555  3.682
## Number of obs: 444, groups: ID, 100
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept) 113.05356   81.03207   1.395
## Year        -0.05498    0.04020  -1.368
##
## Correlation of Fixed Effects:
##      (Intr)
## Year -1.000
```

Again, I see no significant effects.

```
performance::r2(YearwRE_mod)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.157
## Marginal R2: 0.005
```

The marginal R^2 is 0.005. This model is much worse than all the fixed effects included and just pod included as a fixed effect.

4. only julian day as a fixed effect + my random effect.

$$y_i = \beta_0 - \beta_1 \text{julian}_i + \delta_{ID_i} + \epsilon_i \epsilon_i \sim N(0, \sigma_\epsilon^2) \delta_i \sim N(0, \sigma_\delta^2)$$

```
julwRE_mod <- lmer(y_logit ~ julian_day + (1 | ID), data = df_chin)
summary(julwRE_mod)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: y_logit ~ julian_day + (1 | ID)
## Data: df_chin
##
```

```
## REML criterion at convergence: 2377.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.83574 -0.66198 -0.01292  0.87858  1.86197
##
## Random effects:
##  Groups   Name                Variance Std.Dev.
##  ID       (Intercept)    2.496     1.580
##  Residual                  13.144     3.625
## Number of obs: 428, groups: ID, 99
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  4.740098   0.621456   7.627
## julian_day   -0.011497   0.002612  -4.402
##
## Correlation of Fixed Effects:
##              (Intr)
## julian_day  -0.908
```

Similar to previous models in that there are no significant effects here.

```
performance::r2(julwRE_mod)
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.195
##   Marginal R2: 0.043
```

The marginal R^2 is 0.043. This model is slightly worse than all the fixed effects included, but better than the model with just pod and just year.

To use another model selection tool other than R^2 , I will calculate the AICc and Akaike weights to compare these models:

```
## Collect models in a named list
mod_list <- list(
  allwRE_mod = allwRE_mod,
  podwRE_mod = podwRE_mod,
  YearwRE_mod = YearwRE_mod,
  julwRE_mod = julwRE_mod
)

## Compute AICc for each
mod_res <- data.frame(
  model = names(mod_list),
  AICc = sapply(mod_list, MuMIn::AICc)
)

## Order by AICc
mod_res <- mod_res[order(mod_res$AICc), ]
```

```

## deltaAICc
mod_res$delta_AICc <- mod_res$AICc - min(mod_res$AICc)

## Akaike weights based on AICc
mod_res$weight <- exp(-0.5 * mod_res$delta_AICc)
mod_res$weight <- mod_res$weight / sum(mod_res$weight)

## Round for easier viewing
mod_res$AICc <- round(mod_res$AICc, 2)
mod_res$delta_AICc <- round(mod_res$delta_AICc, 2)
mod_res$weight <- round(mod_res$weight, 2)

mod_res

```

```

##           model      AICc delta_AICc weight
## julwRE_mod   julwRE_mod 2385.65      0.00  0.92
## allwRE_mod   allwRE_mod 2390.62      4.97  0.08
## podwRE_mod   podwRE_mod 2474.62     88.97  0.00
## YearwRE_mod  YearwRE_mod 2479.35     93.70  0.00

```

According to the AICc and Akaike weights, the best model here is actually the model that just includes julian day, even though it's R^2 was a bit worse than the model that included all the fixed effects.

In summary, across all LMMs, julian_day had the strongest (though still weak) association with Chinook proportion; Pod and Year explained little additional variation. Overall effect sizes were small, and model diagnostics suggest that a distribution better suited to bounded compositional data may be better.

Moving forward, an LMM with a logit transformation on the y may not be the best model structure for this data. Many of the assumptions of an LMM are probably violated with these data. It may be best to use a GLMM instead. In that case, a beta regression with a logit link may be the most appropriate. Zooming out, I just used percent chinook in the diet here as the y, but there are many other fish detected in this dataset. Because the actual data are compositional, there may be other models, such as multinomial mixture models, rather than forms of linear models, that may be more appropriate for this type of data.

References: 1. Krahn, M. M. (Margaret M. 2004 status review of southern resident killer whales (*Orcinus orca*) under the Endangered Species Act. <https://repository.library.noaa.gov> (2004). 2. Ford, M. J. et al. Status review update for Pacific salmon and steelhead listed under the Endangered Species Act: Pacific Northwest. <https://repository.library.noaa.gov> (2011). 3. Switzer, A. D. et al. Fifteen-year microbiome survey of endangered killer whales (*Orcinus orca*) reveals declining diversity and population differences. 2025.12.27.696689 Preprint at <https://doi.org/10.64898/2025.12.27.696689> (2026). 4. Van Cise, A. M. et al. Spatial and seasonal foraging patterns drive diet differences among north Pacific resident killer whale populations. *Royal Society Open Science* 11, rsos240445 (2024).